

MAT 3150 — Homework 3

Due September 24

Problem 1 (10 points)

Prove that for any $x, y \in \mathbb{R}$,

$$||x| - |y|| \leq |x - y|.$$

Proof.

□

Problem 2 (10 points)

Use the limit definition to compute

$$\lim_{n \rightarrow \infty} \frac{n+2}{3n+5}.$$

Claim. $\lim_{n \rightarrow \infty} \frac{n+2}{3n+5} = L$, where $L = \underline{\hspace{2cm}}$.

Proof.

□

Problem 3 (10 points)

Prove that if $\lim_{n \rightarrow \infty} x_n = \infty$ and $(y_n)_n$ is bounded, then

$$\lim_{n \rightarrow \infty} (x_n + y_n) = \infty.$$

Proof.

□

Problem 4 (10 points)

Let $(x_n)_n$ be an increasing sequence of positive real numbers. Define

$$y_n = \frac{x_1 + x_2 + \cdots + x_n}{n}.$$

Prove that (y_n) is increasing as well.

Proof.

□